Tutorial Notes for Numerical Optimization

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Abstract

The tutorial notes provide addition information of AIST3010.

1 Introduction

In this course, we focus on the following optimization problems

$$\begin{aligned} &\min_{\mathbf{x}} f(\mathbf{x}) \\ &\text{Subject to} \quad \mathbf{x} \in \mathcal{X}, \end{aligned} \tag{1}$$

where $\mathcal{X} \subseteq \mathbb{R}^d$.

2 Matrix Analysis

This section provide some techniques in matrix analysis. To learn more about matrix, please refer to matrix cookbook[1]

We use matrix analysis when

- Calculate and analysis the gradient and Hessian of the object function f
- The constraint on $\mathbf{x}$ can be written as a linear system, and linear algebra helps us do the linear programming tasks..
- ...

2.1 Norm

Definition 2.1. *l_p-norm: for $p \geq 1$*

$$\|\mathbf{x}\|_p = (|x_1|^p + |x_2|^p + \cdots + |x_d|^p)^{\frac{1}{p}} \tag{2}$$

where

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}. \tag{3}$$

[1] http://www2.imm.dtu.dk/pubdb/edoc/imm3274.pdf

Proposition 2.2. $\|\mathbf{x}\|_\infty \overset{def}{=} \lim_{p\to\infty} \|\mathbf{x}\|_p$, *which is equal to*

$$\|\mathbf{x}\|_\infty = \max_{1\le i\le d} |\mathbf{x}_i| \tag{4}$$

Proof. Hint: W.O.L.G, we assume $0 < |x_1| = \max_{1\le i\le d} |x_i|$

$$\|\mathbf{x}\|_\infty = \lim_{p\to\infty} \|\mathbf{x}\|_p = \lim_{p\to\infty} (|x_1|^p + |x_2|^p + \cdots + |x_d|^p)^{\frac{1}{p}} \tag{5}$$

$$= \lim_{p\to\infty} \left(|x_1|^p \left(1 + \frac{|x_2|^p}{|\mathbf{x}_1|^p} + \cdots + \frac{|x_d|^p}{|x_1|^p} \right) \right)^{\frac{1}{p}} \tag{6}$$

$$= |x_1| \lim_{p\to\infty} \left(1 + \frac{|x_2|^p}{|\mathbf{x}_1|^p} + \cdots + \frac{|x_d|^p}{|x_1|^p} \right)^{\frac{1}{p}} \tag{7}$$

Then using squeeze theorem. □

Proposition 2.3.

$$1 \le p \le q \le \infty \Rightarrow \|\mathbf{x}\|_p \ge \|\mathbf{x}\|_q \tag{8}$$

Proof. Hint: Construct the function: $f(r) = (|x_1|^r + \cdots + |x_d|^r)^{1/r}$, and prove that $f'(r) \le 0$ for $r \in [1, \infty]$.[2] □

We then focus on l_1, l_2 and l_∞ norm

Proposition 2.4. *We have*

$$\|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1 \le \sqrt{d}\|\mathbf{x}\|_2 \tag{9}$$

Proof. $\|\mathbf{x}\|_2 \le \|\mathbf{x}\|_1$ can be directly obtained by Proposition 2.3. The right half inequality can be established by using Cauchy–Schwarz inequality that is

$$\|\mathbf{x}\|_1 = 1 \cdot |x_1| + \cdots + 1 \cdot |x_d| \le \sqrt{\underbrace{1 + \cdots + 1}_{d}} \sqrt{\sum_{i=1}^{d} |x_i|^2} = \sqrt{d}\|\mathbf{x}\|_2 \tag{10}$$

□

Questions

1. For every vector $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have $\mathbf{x}^\top \mathbf{y} \le \|\mathbf{x}\|_1 \|\mathbf{y}\|_1$? (True)
2. For every vector $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$, we have $\mathbf{x}^\top \mathbf{y} \le \|\mathbf{x}\|_\infty \|\mathbf{y}\|_\infty$? (False)

[2] For the details, you can check: `https://math.stackexchange.com/questions/1946511`.

2.2 PSD and PD

We only focus on the matrix in the real space.

Definition 2.5. ***Symmetric matrix:*** *$\mathbf{H} = \mathbf{H}^\top$, we call $\mathbf{H}$ is symmetric.*

Proposition 2.6. *For every real symmetric matrix $\mathbf{H}$ and $\operatorname{rank}(\mathbf{H}) = r$, there exists an orthogonal matrix $\Gamma \in \mathbb{R}^{r\times d}$ where $\Gamma^\top = [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_r]$ and a diagonal matrix $\Sigma = \operatorname{diag}(\lambda_1; \cdots; \lambda_r) \in \mathbb{R}^{r\times r}$ such that*

$$\mathbf{H} = \Gamma^\top \Sigma \Gamma \tag{11}$$

Remark 2.7. *We can always find an orthogonal matrix $\bar{\Gamma} \in \mathbb{R}^{d\times d}$ s.t. $\bar{\Gamma}^\top = [\Gamma^\top, \mathbf{y}_{r+1}, \cdots, \mathbf{y}_d]$ and*

$$\mathbf{H} = \bar{\Gamma}^\top \bar{\Sigma} \bar{\Gamma}, \tag{12}$$

where $\bar{\Sigma} = \operatorname{diag}(\lambda_1; \cdots; \lambda_r; 0; \cdots; 0) \in \mathbb{R}^{d\times d}$. We always use this type of decomposition when doing matrix analysis.

Definition 2.8. ***Postive-(Semi)Definite Matrix:*** *$\mathbf{H} \in \mathbb{R}^{d\times d}$ is symmetric, we call $\mathbf{H}$ positive-semidefinite ($\mathbf{H} \succeq \mathbf{0}$) if:*

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} \geq 0, \quad \textit{for all} \;\; \mathbf{x} \neq \mathbf{0}, \;\; \mathbf{x} \in \mathbb{R}^d. \tag{13}$$

We call $\mathbf{H}$ positive-definite ($\mathbf{H} \succ \mathbf{0}$) if

$$\mathbf{x}^\top \mathbf{H} \mathbf{x} > 0, \quad \textit{for all} \;\; \mathbf{x} \neq \mathbf{0}, \;\; \mathbf{x} \in \mathbb{R}^d. \tag{14}$$

Proposition 2.9. *If $\mathbf{H}$ is a symmetric matrix, then the following statements*

- *(a).* $\mathbf{H} \succeq \mathbf{0}$
- *(b).* $\mathbf{H}$*'s eigenvalues are all non-negative*
- *(c). There exits a matrix $\mathbf{A}$ s.t. $\mathbf{H} = \mathbf{A}^\top \mathbf{A}$*

Proof. We can prove this by the loop

$$\text{(c)} \Rightarrow \text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \tag{15}$$

□

Proposition 2.10. *If $\mathbf{H}$ is a symmetric matrix, then the following statements*

- *(a).* $\mathbf{H} \succ \mathbf{0}$
- *(b).* $\mathbf{H}$*'s eigenvalues greater than 0*
- *(c). There exits a matrix $\mathbf{A} \succ \mathbf{0}$ s.t. $\mathbf{H} = \mathbf{A}^\top \mathbf{A}$*

Proof. We can prove this by the loop

$$\text{(c)} \Rightarrow \text{(a)} \Rightarrow \text{(b)} \Rightarrow \text{(c)} \tag{16}$$

□

Questions

1. Every PSD matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ is full-rank? (False)
2. Every PD matrix $\mathbf{A} \in \mathbb{R}^{d\times d}$ is full-rank? (True)
3. If $\mathbf{A} \succ \mathbf{0}$, then $\mathbf{A}^{-1} \succ \mathbf{0}$ (True)
4. If $\mathbf{A} \succ \mathbf{B} \succ \mathbf{0}$, then $\mathbf{A}^2 \succ \mathbf{B}^2$ (False)*
5. If $\mathbf{A} = \begin{bmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{bmatrix} \succ \mathbf{0}$, then $\mathbf{A}_{11} \succ \mathbf{0}$ and $\mathbf{A}_{22} \succ \mathbf{0}$ (True)

2.3 Matrix Derivative

Consider a function $f(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}^1$, the gradient of f is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_d} \end{bmatrix} \in \mathbb{R}^d. \tag{17}$$

Consider a function $\mathbf{F}(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix} : \mathbb{R}^d \to \mathbb{R}^m$, The derivative/Jacobi Matrix of $\mathbf{F}$ is

$$d_{\mathbf{F}}(\mathbf{x}) = \mathbf{J}_{\mathbf{F}}(\mathbf{x}) = \frac{\partial \mathbf{F}(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \nabla F_1(\mathbf{x})^\top \\ \nabla F_2(\mathbf{x})^\top \\ \vdots \\ \nabla F_m(\mathbf{x})^\top \end{bmatrix} \in \mathbb{R}^{m\times d} \tag{18}$$

If we choose $\mathbf{F}(\mathbf{x}) = \nabla f(\mathbf{x}) : \mathbb{R}^d \to \mathbb{R}^d$ ($F_i(\mathbf{x}) = \nabla_{x_i} f(\mathbf{x})$), the Hessian matrix of f is defined to be

$$\nabla^2 f(\mathbf{x}) = \mathbf{J}_{\nabla f}(\mathbf{x}) = \left[\frac{\partial^2 f(\mathbf{x})}{\partial x_i \partial x_j}\right]_{ij} \in \mathbb{R}^{d\times d} \tag{19}$$

Rules for calculating the derivative (these rules can be treated as a generation for the rule in Single Variable Calculus

- $d_{\mathbf{h}^\top \mathbf{g}}(\mathbf{x}) = \mathbf{h}(\mathbf{x})^\top d_{\mathbf{g}}(\mathbf{x}) + \mathbf{g}(\mathbf{x})^\top d_{\mathbf{h}}(\mathbf{x})$
- (Chain's Rule) $d_{\mathbf{h}\circ\mathbf{g}}(\mathbf{x}) = d_{\mathbf{h}}(\mathbf{g}(\mathbf{x}))d_{\mathbf{g}}(\mathbf{x})$

Questions

1. $\mathbf{F}(\mathbf{x}) = \mathbf{A}\mathbf{x}$: $\frac{\partial \mathbf{F}(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}$
2. $f(\mathbf{x}) = \mathbf{a}'\mathbf{x}$: $\nabla f(\mathbf{x}) = \mathbf{a}$ and $\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = [\nabla f(\mathbf{x})^\top] = \mathbf{a}'$

3. $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x}$:

$$d_f(\mathbf{x}) = \frac{\partial(\mathbf{A}\mathbf{x})^\top \mathbf{x}}{\partial \mathbf{x}} = (\mathbf{A}\mathbf{x})^\top \frac{\partial \mathbf{x}}{\partial \mathbf{x}} + \mathbf{x}^\top \frac{\partial(\mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = \mathbf{x}^\top \mathbf{A}^\top + \mathbf{x}^\top \mathbf{A} \tag{20}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} \tag{21}$$

$$\nabla^2 f(\mathbf{x}) = d_{\nabla f}(\mathbf{x}) = \mathbf{A} + \mathbf{A}^\top \tag{22}$$

If $\mathbf{A}$ is symmetric, then $\nabla f(\mathbf{x}) = 2\mathbf{A}\mathbf{x}$

4. $f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2$ (we let $\mathbf{u} = \mathbf{A}\mathbf{x} - \mathbf{b}$)

$$d_f(\mathbf{x}) = \frac{\partial \|\mathbf{u}\|^2}{\partial \mathbf{u}} \frac{\partial \mathbf{u}}{\partial \mathbf{x}} = 2\mathbf{u}'\mathbf{A} = 2(\mathbf{A}\mathbf{x} - \mathbf{b})^\top \mathbf{A} \tag{23}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = 2\mathbf{A}^\top(\mathbf{A}\mathbf{x} - \mathbf{b}) \tag{24}$$

$$\nabla^2 f(\mathbf{x}) = d_{\nabla f}(\mathbf{x}) = 2\mathbf{A}^\top \mathbf{A} \tag{25}$$

5. $f(\mathbf{x}) = \log\left(1 + \mathbf{e}^{-\mathbf{a}'\mathbf{x}}\right)$:

$$d_f(\mathbf{x}) = -\frac{1}{1 + \mathbf{e}^{-\mathbf{a}'\mathbf{x}}} \mathbf{e}^{-\mathbf{a}'\mathbf{x}} \mathbf{a}' \tag{26}$$

$$\nabla f(\mathbf{x}) = [d_f(\mathbf{x})]^\top = -\left(\frac{\mathbf{e}^{-\mathbf{a}'\mathbf{x}}}{1 + \mathbf{e}^{-\mathbf{a}'\mathbf{x}}}\right)\mathbf{a} \tag{27}$$

$$\nabla^2 f(\mathbf{x}) = p(1-p)\mathbf{a}\mathbf{a}' \tag{28}$$

where $p = \left(\frac{1}{1+\mathbf{e}^{-\mathbf{a}'\mathbf{x}}}\right)$.

2.4 Matrix Integral

The following position will be useful to obtain the properties of a convex function, you can skip them first.

Proposition 2.11.

$$f(\mathbf{y}) - f(\mathbf{x}) = \int_{t=0}^{1} \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))^\top (\mathbf{y} - \mathbf{x}) dt \tag{29}$$

$$\nabla f(\mathbf{y}) - \nabla f(\mathbf{x}) = \int_{t=0}^{1} \nabla^2 f(\mathbf{x} + t(\mathbf{y} - \mathbf{x}))(\mathbf{y} - \mathbf{x}) dt \tag{30}$$

3 Convex Function

Definition 3.1. *(**Convex function**) f is a convex function if*

$$f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) \leq \theta f(\mathbf{x}) + (1-\theta) f(\mathbf{y}) \tag{31}$$

for all $\theta \in [0,1]$ and $\mathbf{x}, \mathbf{y} \in \mathrm{dom}(f)$

Proposition 3.2. *If f is twice differentiable, the following statements*

- *(a). f is convex*

- *(b).* $f(\mathbf{y}) \geq f(\mathbf{x}) + \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle$
- *(c).* $\nabla^2 f(\mathbf{x}) \succeq \mathbf{0}$

are equal

Proof. We first prove $(a) \Leftrightarrow (b)$.
(a) $\Rightarrow$ (b): for all $\theta \in [0, 1]$ and $\mathbf{x}_\theta = \theta\mathbf{x} + (1-\theta)\mathbf{y}$, we have for all θ

$$f(\mathbf{y}) \geq \frac{1}{1-\theta}(f(\mathbf{x}_\theta) - \theta f(\mathbf{x})) = f(\mathbf{x}) + \frac{f(\mathbf{x}_\theta) - f(\mathbf{x})}{1-\theta} \tag{32}$$

we then have

$$f(\mathbf{y}) - f(\mathbf{x}) \geq \lim_{\theta \to 1} \frac{f(\theta\mathbf{x} + (1-\theta)\mathbf{y}) - f(\mathbf{x})}{1-\theta} = \lim_{\theta \to 1} \frac{f(\mathbf{x} + (1-\theta)(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{1-\theta} \tag{33}$$

$$= \lim_{\alpha \to 0} \frac{f(\mathbf{x} + \alpha(\mathbf{y} - \mathbf{x})) - f(\mathbf{x})}{\alpha} = \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \tag{34}$$

(b) $\Rightarrow$ (a): we have

$$f(\mathbf{x}_\theta) \leq f(\mathbf{y}) - \theta \langle \nabla f(\mathbf{x}_\theta), \mathbf{y} - \mathbf{x} \rangle \tag{35}$$

$$f(\mathbf{x}_\theta) \leq f(\mathbf{x}) + (1-\theta) \langle \nabla f(\mathbf{x}_\theta), \mathbf{y} - \mathbf{x} \rangle \tag{36}$$

We then prove $(b) \Leftrightarrow (c)$.
(b) $\Rightarrow$ (c) for all $\mathbf{y}$, we have

$$f(\mathbf{y}) - f(\mathbf{x}) \geq \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle \tag{37}$$

Please refer to Theorem 2.1.4 in Nesterov's book[3]
(c) $\Rightarrow$ (b)

$$f(\mathbf{y}) - f(\mathbf{x}) - \langle \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle = \int_{t=0}^{1} \langle \nabla f(\mathbf{x} + t(\mathbf{y} - \mathbf{x})) - \nabla f(\mathbf{x}), \mathbf{y} - \mathbf{x} \rangle dt \tag{38}$$

$$= \int_{t=0}^{1} \langle \int_{s=0}^{1} \nabla^2 f(\mathbf{x} + ts(\mathbf{y} - \mathbf{x}))t(\mathbf{y} - \mathbf{x}) ds, \mathbf{y} - \mathbf{x} \rangle dt \tag{39}$$

$$= \int_{t=0}^{1} t(\mathbf{y} - \mathbf{x})^\top \left(\int_{s=0}^{1} \nabla^2 f(\mathbf{x} + ts(\mathbf{y} - \mathbf{x})) ds \right) (\mathbf{y} - \mathbf{x}) dt \geq 0 \tag{40}$$

□

[3] `https://link.springer.com/book/10.1007/978-3-319-91578-4`